

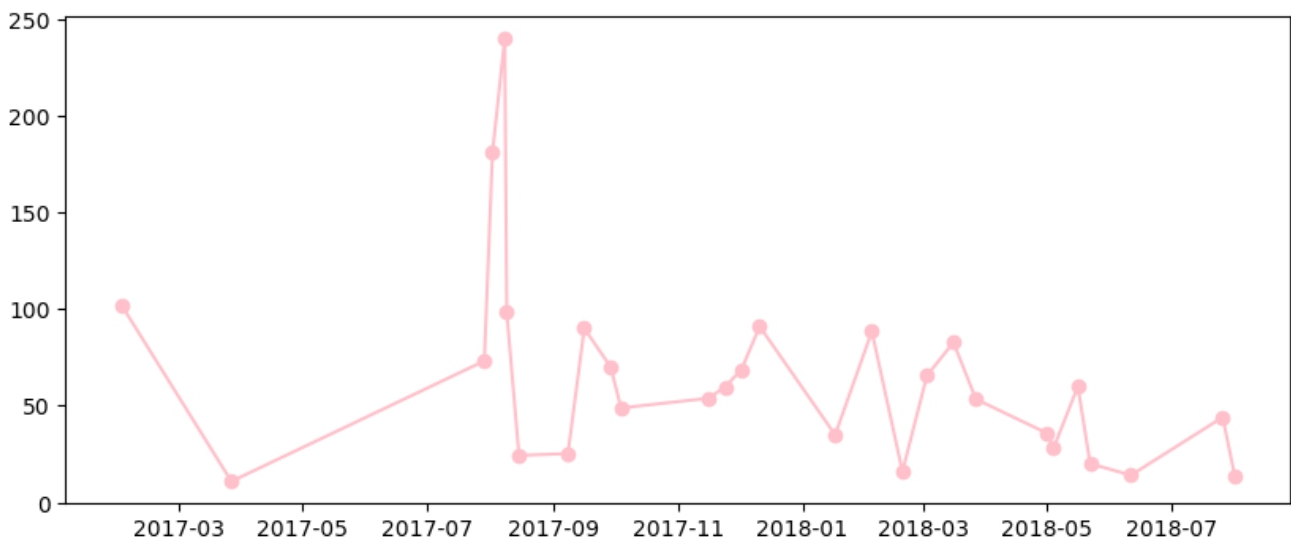
Visualization Assessment Solution Report

Question 1: 1. Line Chart (Trend)

Your Code:

```
daily_fares = merged_df.groupby(merged_df['ride_date'].dt.date)['fare_amt'].sum().reset_index()
daily_fares
# Write your plotting code here:
plt.figure(figsize=(10,4))
plt.plot(daily_fares['ride_date'],daily_fares['fare_amt'],color='pink',linestyle='solid',marker='o')
plt.show()
# ...
```

Rendered Output:



Question 2: 2. Bar Chart (Categorical)

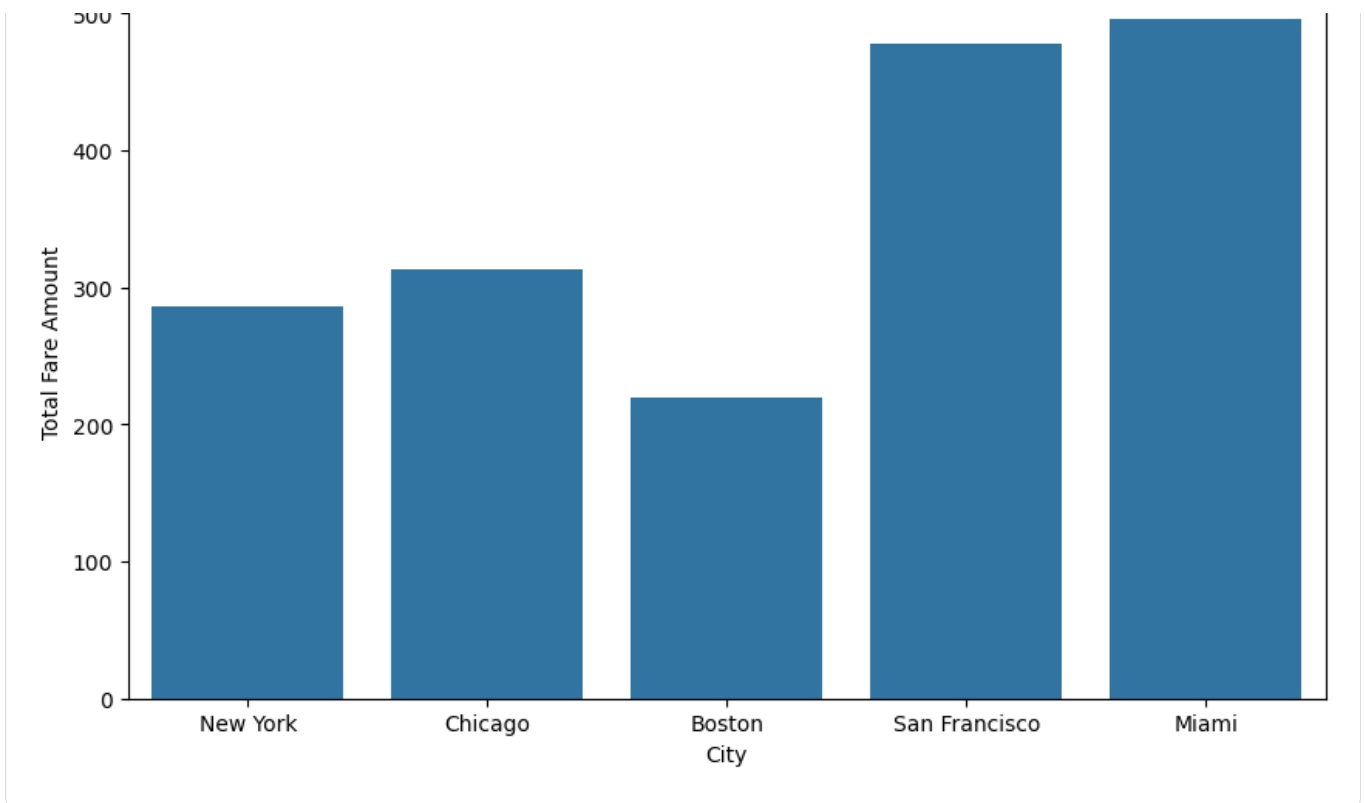
Your Code:

```
plt.figure(figsize=(10, 6))
sns.barplot(data=merged_df,x="city",y="fare_amt",estimator=sum,errorbar=None)

plt.title("Total Fare Amount by City")
plt.xlabel("City")
plt.ylabel("Total Fare Amount")
#
plt.show()
```

Rendered Output:

Total Fare Amount by City



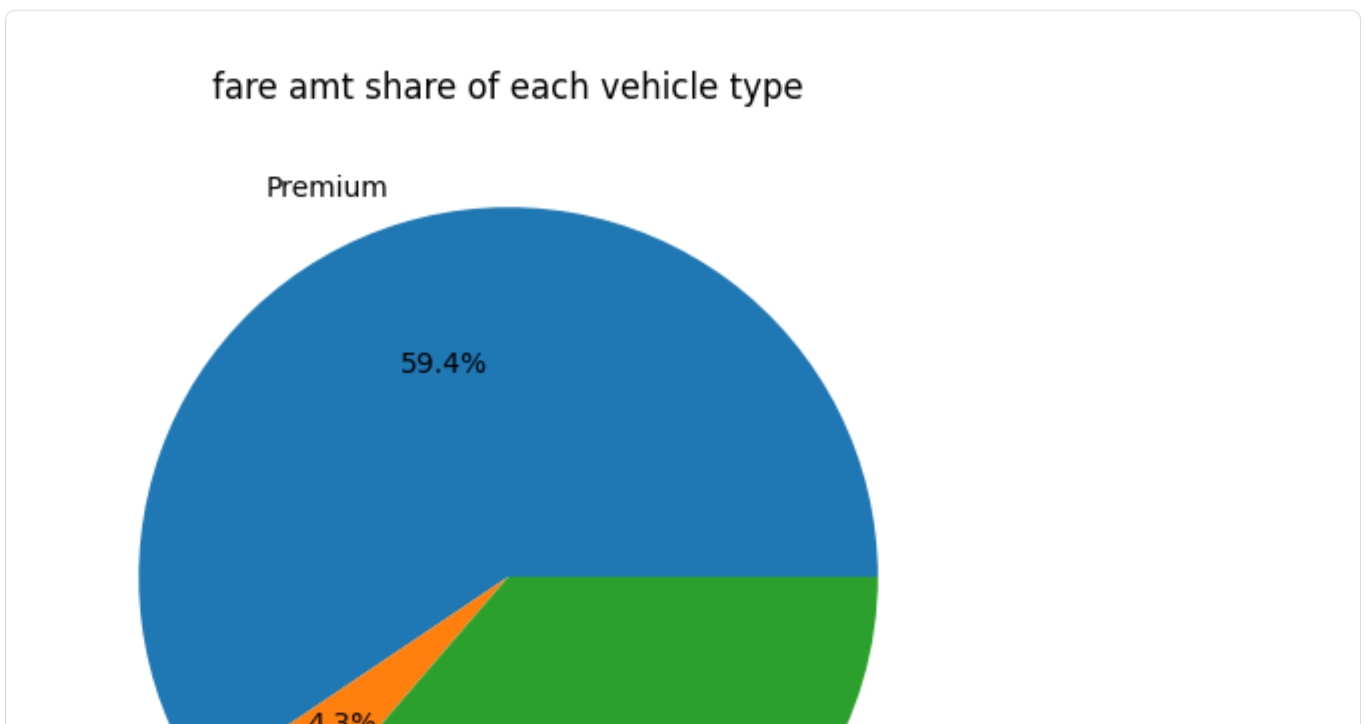
Question 3: 3. Pie Chart (Distribution)

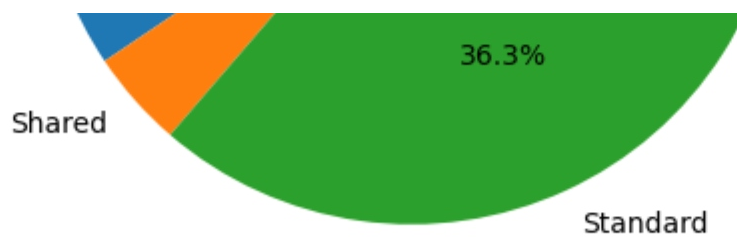
Your Code:

```
vehicle_fares = merged_df.groupby('vehicle_type')['fare_amt'].sum().reset_index()

# Write your plotting code here:
plt.pie(vehicle_fares['fare_amt'], labels=vehicle_fares['vehicle_type'], autopct='%1.1f%%')
plt.title("fare amt share of each vehicle type")
plt.show()
```

Rendered Output:



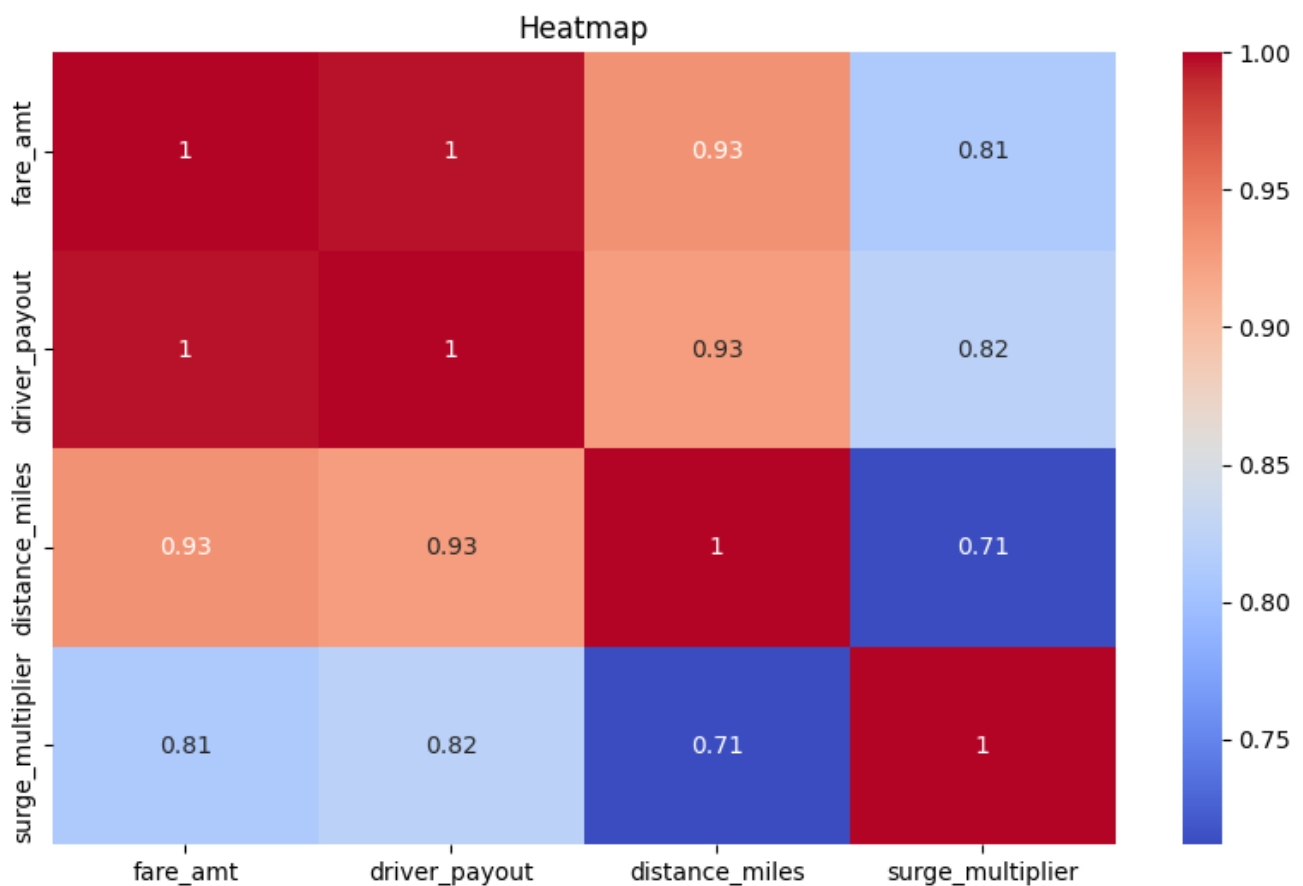


Question 4: 4. Heatmap (Correlation)

Your Code:

```
corr_cols = ['fare_amt', 'driver_payout', 'distance_miles', 'surge_multiplier']
# Write your plotting code here:
corr_matrix = merged_df[corr_cols].corr()
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm')
plt.title("Heatmap")
plt.show()
```

Rendered Output:



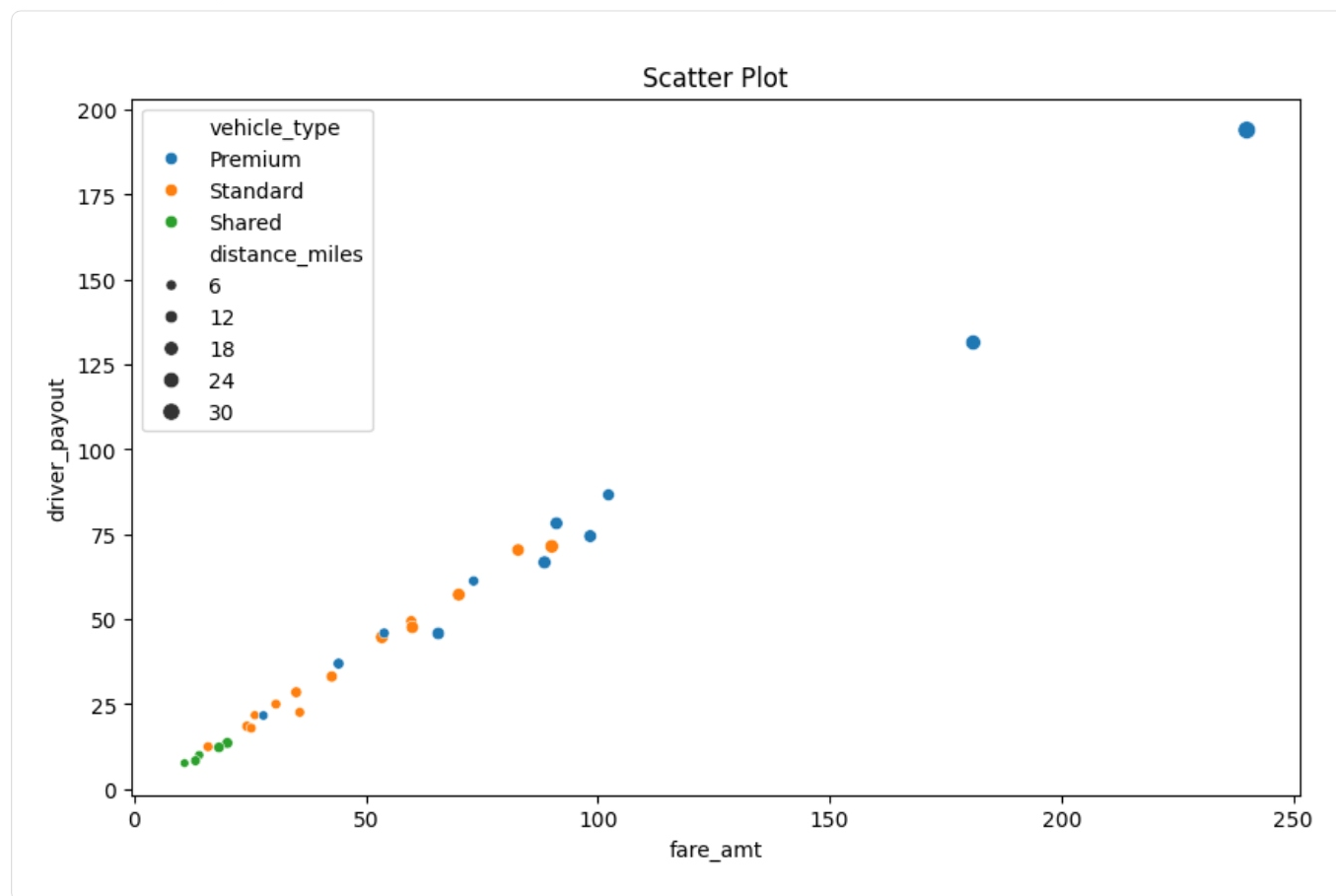
Question 5: 5. Scatter Plot (Relationships)

Your Code:

```
sns.scatterplot(data=merged_df, x='fare_amt', y='driver_payout', hue='vehicle_type', size='dista
```

```
plt.title("Scatter Plot")
plt.show()
```

Rendered Output:



Question 6: 6. Interactive Histogram

Your Code:

```
import matplotlib.pyplot as plt
plt.hist(merged_df["driver_payout"], bins=15)
plt.show()
```

Rendered Output:

